1. What is API testing?

API testing is basically testing backend services without UI. I verify correct data, status codes, and performance. For example, I tested login API to ensure it returns a token for valid users.

2. What is REST API?

REST API is a standard way to build APIs using HTTP methods. It is stateless and usually returns JSON. For example, GET /users/1 returns user details.

3. REST vs SOAP?

REST is lightweight, uses JSON, and is faster. SOAP uses XML and is more secure but heavier. In my projects, we used REST APIs.

4. What are HTTP methods?

HTTP methods define actions like GET, POST, PUT, DELETE, PATCH. For example, POST is used to create a user.

5. Explain GET request.

GET is used to fetch data. For example, GET /users returns list of users.

6. Explain POST request.

POST is used to create resources. For example, POST /users creates a new user with JSON data.

7. Explain PUT request.

PUT is used to update an entire resource. For example, PUT /users/1 replaces user data.

8. Explain DELETE request.

DELETE removes a resource. Example: DELETE /users/1 deletes user.

9. What is PATCH request?

PATCH is used for partial updates like updating only email field.

10. What are status codes?

Status codes indicate API response results like success or failure.

11. Explain 200 status code.

200 means success.

12. Explain 201 status code.

201 means resource created.

13. Explain 400 status code.

400 means bad request due to invalid input.

14. Explain 401 status code.

401 means authentication failed.

15. Explain 403 status code.

403 means access denied.

16. Explain 404 status code.

404 means resource not found.

17. Explain 500 status code.

500 means server error.

18. What is Postman?

Postman is a tool I used for API testing where I create requests, add scripts, and validate responses.

19. What is Newman CLI?

Newman is a CLI tool to run Postman collections in pipelines.

20. How did you integrate Newman with GitHub Actions?

I created a YAML pipeline where newman run command executes automatically on code push.

21. What are headers in API?

Headers are metadata like Content-Type and Authorization.

22. What is authorization?

Authorization defines what access a user has.

23. What is authentication?

Authentication verifies user identity.

24. Difference between authentication and authorization?

Authentication is who you are, authorization is what you can access.

25. What is Bearer Token?

Bearer token is used for API authentication and passed in Authorization header.

26. What is mTLS authentication?

mTLS ensures both client and server verify certificates.

27. Explain certificate-based authentication.

It uses certificates instead of username/password for secure access.

28. How did you validate TLS versions?

I used OpenSSL to check support for TLS 1.2 and above.

29. How did you validate cipher suites?

Used OpenSSL commands to verify strong encryption algorithms.

30. How did you use OpenSSL in testing?

Used it for checking certificates, TLS handshake, and encryption settings.

31. What is payload validation?

Validating request/response data correctness.

32. What is schema validation?

Validating JSON structure using schema.

33. What is API chaining?

Using one API response data in another API.

34. How did you validate KMS APIs?

Tested key creation, encryption, and decryption functionality.

35. How did you validate LMS APIs?

Validated user enrollment and course APIs.

36. What is DRM workflow testing?

Testing content security, license validation and playback.

37. How did you test license generation APIs?

Ensured valid users get licenses.

38. How did you test license expiry?

Verified expired licenses block access.

39. What is GeoIP validation?

Ensures access based on user location.

40. How did you validate User-Agent headers?

Tested API responses for mobile vs desktop agents.

41. How do you validate response time?

Ensured APIs meet SLA like <200ms.

42. What is negative testing in APIs?

Testing with invalid inputs to verify proper errors.

43. How do you handle dynamic tokens?

Extract token and reuse using Postman scripts.

44. What are query parameters?

Optional filters like /users?role=admin.

45. What are path parameters?

Identifiers like /users/1.

46. What are environment variables in Postman?

Used to store reusable values like base URL and token.

47. Explain your Node.js API framework.

Used Axios for API calls and Mocha/Chai for validation.

48. How did you reduce manual effort by 60%?

Automated regression tests and integrated CI/CD.

49. Explain end-to-end API validation workflow.

Login, get token, create resource, validate, delete.

50. Explain your API testing project architecture.

Used layered structure with test, utility, and config integrated with CI/CD.